NEXORA

AI SYNERGY HACKATHON 2026

SOFTWARE REQUIREMENTS SPECIFICATION

Uncertainty-Aware Food Nutrition Estimator

TEAM	Nexora
TEAM LEADER	Sidharth Bhadauria
TEAM MEMBERS	Tanu Soni • Ramakant Sharma • Priyanka
INSTITUTION	Amity University Madhya Pradesh
DOCUMENT STATUS	Enhanced hackathon SRS — proposed design and roadmap

A transparent nutrition-estimation workflow that communicates uncertainty instead of implying false precision.

1. Executive Summary

Nexora proposes an uncertainty-aware, photo-based food nutrition estimator for the AI Synergy Hackathon 2026. The planned system recognises visible foods, estimates portions, predicts calories and macronutrients, and communicates how reliable each estimate is. When a photograph does not provide enough evidence, the system should widen the estimate range or ask a short follow-up question instead of presenting false precision.

Design principle: when evidence is weak, widen the range or ask a question—never invent precision.

2. Problem Statement and Objectives

A meal photograph does not directly reveal exact food weight, hidden ingredients, oil or ghee quantity, or cooking method. A single calorie value can therefore imply more certainty than the image supports.

- Recognise visible food items and identify uncertain labels.
- Estimate portion size and nutrition with ranges where evidence supports them.
- Use confidence checks to decide whether to show a result, show a caution, or request clarification.
- Refine estimates when users provide dish, portion, or added-oil details.
- Evaluate the prototype transparently, without claiming unmeasured accuracy.

3. Scope, Users, and Boundaries

The proposed hackathon prototype covers one uploaded RGB meal photograph (JPG/PNG), optional text details, food-item recognition, portion estimation, calories, protein, carbohydrates and fat. It is intended for general dietary tracking only, not diagnosis, treatment, or personalised medical advice.

User	Need / success outcome
Everyday user	Quick estimate with understandable ranges and simple prompts.
Diet-tracking user	Per-item and meal-total nutrition ranges with confidence status.
Reviewer / researcher	Repeatable tests, a baseline comparison, and documented limitations.

Out of scope for the first version: real-time video, multi-photo reconstruction, micronutrients, disease-specific diet plans, barcode scanning, reliable automatic hidden-oil detection, training a large dataset from scratch, user accounts, long-term history, and a dedicated mobile app.

4. Inputs and Outputs

Type	Specification
Required input	One RGB meal photograph in JPG or PNG format.
Optional user details	Dish name, ingredient hints, portion size (e.g., “2 rotis”), and added oil or ghee.
System-derived cues	Food labels, segmentation masks, and an estimated depth map where available.
Food output	Detected item names; uncertain labels visibly flagged.
Portion output	Grams or pieces, with a range where supported.
Nutrition output	Calories, protein, carbohydrates and fat per item and for the meal total, with ranges where supported.
Confidence output	“Estimate shown”, “Wide range — treat with caution”, or a clarifying question.
Next step	Explain which detail could improve the estimate, such as portion size or added oil.

Illustrative example only: “Rice 180 ± 15 g; total energy 720 ± 50 kcal.” These are example values, not measured results or a performance claim.

Every result screen should show: “For general dietary tracking only; not medical advice.”

5. System Architecture

The architecture separates image understanding, portion estimation, nutrition prediction and uncertainty handling. The seven stages below are laid out in two rows so the diagram remains within the page margins.

1. MEAL PHOTO
RGB JPG / PNG
→
2. IMAGE PROCESSING
Validate quality
→
3. FOOD RECOGNITION
Identify / segment
→
4. PORTION ESTIMATION
Estimate quantity

↓ **Low confidence: ask about portion size / dish / added oil or ghee, then re-estimate.**

5. NUTRITION PREDICTION
Calories + macros
→
6. CONFIDENCE CHECK
Assess uncertainty
→
7. FINAL RESULT
Range / warning

Low-confidence loop: if the evidence is insufficient, ask about portion size, dish identity, or added oil/ghee. Use the answer to update the estimate and confidence status. If uncertainty remains, preserve a wider range and explain the limitation.

• Image preparation checks blur, darkness, file validity and whether the image appears to contain food.

• Recognition and segmentation identify visible foods and image regions; segmentation is used only where supported by the selected model.

• Portion estimation combines appearance with estimated depth cues when available. These are predictions, not direct measurements.

• Nutrition prediction uses an available nutrition dataset/reference and a mapping from food and portion to nutrient estimates.

• Confidence logic chooses between result, caution, and clarification; it must not manufacture confidence scores unsupported by validation.

6. Uncertainty-Aware AI Behaviour

Situation	Expected behaviour
Clear image and familiar food	Return the estimate and its range, with a confidence status.
Blurry, dark, or non-food image	Reject with an understandable reason and ask for a clearer photo.
Unrecognised dish	Mark the label uncertain, widen the range, and ask for the dish name if useful.
Overlapping foods	Lower confidence for affected items and flag uncertain portions.
Depth estimate unavailable	Fall back to RGB/semantic cues and reduce confidence.
Portion unclear	Ask a concise question, e.g., “Was this one bowl or two?”
Oil/ghee not visible	Ask whether oil/ghee was added and, when useful, offer a simple quantity prompt.
User answers a prompt	Recalculate using the supplied detail and show that the estimate was refined.
Model/server error	Show a clear error and retry option; never fabricate a result.

The system should distinguish uncertainty caused by image quality, food identity, portion size, and hidden ingredients. A confidence label is meaningful only if its decision rules are documented and evaluated.

7. Functional Requirements

ID	Requirement
FR-01	Accept one JPG/PNG meal photo and optional text details.
FR-02	Validate image format and basic quality; explain rejection reasons.
FR-03	Identify visible food items and segment them where the selected model supports it.
FR-04	Estimate portions in grams or pieces and flag weak estimates.
FR-05	Use RGB, estimated depth and semantic cues where available; reduce reliance on unreliable or missing cues.
FR-06	Estimate calories, protein, carbohydrates and fat per item and for the meal total.
FR-07	Attach uncertainty ranges to portions and nutrients where supported; clearly mark illustrative or unavailable ranges.
FR-08	Choose a result, caution, or follow-up question using documented confidence rules.
FR-09	Ask about portion, dish identity, or added oil/ghee when relevant; re-estimate after user input.
FR-10	Show the general dietary-use disclaimer on every result screen.
FR-11	Handle invalid input and model/server failures without inventing values.
FR-12	Avoid retaining meal photos beyond the session unless the user explicitly agrees.

Priority for a hackathon demo: complete the end-to-end path first; advanced segmentation, depth estimation, and calibrated uncertainty remain dependent on model availability and time.

8. Proposed Technology Stack

Component	Planned technology	Purpose / status
Frontend	Streamlit	Proposed web demo for upload, results and follow-up questions.
Backend	Python	Connect the image pipeline, nutrition logic and UI.
Image processing	OpenCV	Image loading and basic quality checks.
Food recognition	Compatible pretrained computer-vision model	Recognise foods; segmentation only if supported by chosen model.
AI/ML framework	PyTorch	Run or adapt compatible pretrained models.
Nutrition data	Nutrition5k and/or available validated nutrition references	Support estimation and evaluation; coverage limitations must be documented.
Visualisation	Matplotlib	Plot evaluation results and uncertainty calibration.

This is a planned stack, not a claim that all components are already implemented. Confirm package/model compatibility and dataset licensing before integration.

9. Research Alignment and Technical Rationale

The supplied SRS describes a multimodal, uncertainty-aware approach using RGB appearance, semantic food cues and estimated depth where available. It also states that depth and semantic cues are predictions rather than direct measurements, and that unreliable cues should receive less weight.

• Multimodal evidence: combine available visual and semantic signals rather than treating one cue as definitive.

• Uncertainty-aware output: report ranges and confidence status; ask for missing information when that can materially change the estimate.

• Human-in-the-loop refinement: use user corrections about dish, portion and oil/ghee to update the result.

• Evaluation discipline: compare a multimodal approach with an RGB-only baseline and report measured results only.

• Dataset limitations: Nutrition5k is proposed as a starting point; the supplied SRS notes that its coverage is largely US cafeteria meals and may not represent all cuisines.

Research-paper note: no separate research paper was attached with the supplied SRS. This section is therefore aligned to the uncertainty-aware multimodal method described in the SRS, not to an independently verified paper. Add the paper's exact citation and any paper-specific methods after the paper is supplied.

10. Evaluation Plan and Criteria

Area	Method / reporting requirement
Food recognition	Record correct/incorrect item labels on a held-out set; document sample size and label source.
Portion and calorie error	Report MAE for calories against labelled references; add portion error metrics if reliable ground truth is available.
Baseline comparison	Compare multimodal cues against an RGB-only baseline using the same held-out examples.
Uncertainty quality	Use a reliability diagram and a suitable calibration metric such as ECE only where confidence outputs and labels permit.
Cuisine coverage	Where suitable labelled examples exist, test Indian, East Asian and Western meals; report sample counts per group.
Robustness	Test blurry/dark photos, overlapping items, unfamiliar dishes, missing depth and model/server failures.
Reporting rule	Publish measured results only after experiments. Label targets as targets and include limitations.

No model accuracy, latency, or nutrition-error performance is claimed in this SRS. Such values must be measured on the implemented prototype.

11. Implementation Roadmap

Phase	Work	Deliverable
1. Setup	Confirm scope, inspect datasets, set up environment and repository.	Runnable project skeleton and data plan.
2. Image understanding	Implement photo upload, quality checks and pretrained food recognition.	Demo identifies visible items on selected examples.
3. Nutrition baseline	Connect portion-estimation approach and nutrition mapping.	Per-item and total estimates with stated limitations.
4. Uncertainty loop	Add confidence rules, warnings, portion/oil prompts and re-estimation.	Low-confidence example triggers a question and revised estimate.
5. Evaluation and demo	Run cross-cuisine tests, compare baseline and review errors.	Demo, test cases and evidence-based results summary.

Recommended sequencing: deliver a simple, reliable end-to-end baseline before attempting advanced depth estimation or confidence calibration.

12. Acceptance Criteria

ID	Pass/fail check
AC-01	A clear supported photo returns food items, portions and nutrition with ranges where supported.
AC-02	Every result has a confidence status; unsupported exact-looking values are avoided.
AC-03	An unusable photo is rejected with an understandable reason.
AC-04	Low confidence triggers a relevant question, and user input changes the next estimate or its uncertainty.
AC-05	Calorie MAE is reported against an RGB-only baseline where labelled data permits; improvement is a target to verify, not an assumption.
AC-06	Calibration is evaluated with ECE/reliability diagram where appropriate; no results are invented.
AC-07	The disclaimer appears on every result screen.
AC-08	Test at least one Indian, one East Asian and one Western meal where suitable examples are available, documenting sample limitations.
AC-09	Invalid inputs and model/server errors show safe error states and do not fabricate nutrition values.
AC-10	Photo-retention behaviour follows the stated privacy rule.

Acceptance status should be marked only after each criterion is tested on the actual implementation.

13. Risks, Constraints and Mitigations

Risk / constraint	Mitigation
Hidden oil, ghee, sauces or covered ingredients	Ask the user and widen the range if unresolved.
A 2D image cannot directly reveal exact weight or volume	Treat depth as estimated; report portion uncertainty.
Recipe variation changes nutrition	Use ranges and request cooking/oil details.
Unfamiliar cuisines or dishes	Flag uncertainty and report cuisine-wise results where sample sizes allow.
Limited hackathon time or compute	Use compatible pretrained components and prioritise an end-to-end prototype.
Small or unrepresentative test set	Report sample size and limitations; avoid broad claims.
Privacy	Do not retain meal photos beyond the session unless the user explicitly agrees.
Medical misuse	Display a disclaimer; do not present the prototype as a medical device.

Fallback behaviour: if depth, segmentation, nutrition references or a model service is unavailable, use a clearly labelled reduced-capability path or report that the estimate cannot be produced.

14. Team and Governance

Team member	Proposed responsibility
Sidharth Bhadauria — Team Leader	Coordination, integration tracking and final demo.
Tanu Soni	Dataset research, test-case organisation and evaluation documentation.
Ramakant Sharma	AI/ML exploration, image processing and portion-estimation workflow.
Priyanka	Frontend support, user-flow testing and documentation.

Team: Nexora | Institution: Amity University Madhya Pradesh | Event: AI Synergy Hackathon 2026

Responsibilities are proposed allocations from the supplied SRS and should be updated if the team's actual division of work differs.

15. Privacy, Limitations and Disclaimer

Meal photos may reveal personal routines or surroundings. The prototype should not retain images beyond the session unless the user explicitly agrees. Any storage, logging, or third-party model processing should be disclosed clearly.

The estimate can be affected by occlusion, unusual recipes, portion ambiguity, hidden ingredients, dataset mismatch and model errors. A follow-up question can reduce some uncertainty but cannot make every estimate exact.

DISCLAIMER — This prototype supports general dietary tracking only. It does not provide medical advice and does not replace a qualified doctor or dietitian.

Appendix A. Demo Checklist

- Upload a supported clear photo and verify the normal result path.
- Test a blurry/dark or non-food image and verify an understandable rejection.
- Test an unfamiliar or overlapping dish and verify uncertainty is communicated.
- Trigger a portion-size question and verify that the next estimate changes appropriately.
- Trigger an oil/ghee question and verify the answer is reflected in the estimate or range.
- Verify the disclaimer, error handling, and photo-retention behaviour.
- Record dataset split, sample counts, measured metrics and known limitations before presenting results.